In [11]:
```
from google.colab import files
uploaded = files.upload()
```

Upload widget is only available when the cell has been executed in the current browser session. Please rerun this cell to enable.

```
Saving Social_Network_Ads.csv to Social_Network_Ads (3).csv
```

In [12]:
```
import numpy as np
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import classification_report

df = pd.read_csv('Social_Network_Ads.csv')
print(df)
print(df.head())

features = df.iloc[:, [2, 3]].values
label = df.iloc[:, 4].values

print(features)
print(label)

for i in range(1, 401):
    x_train, x_test, y_train, y_test = train_test_split(
        features, label, test_size=0.2, random_state=i
    )
    model = LogisticRegression()
    model.fit(x_train, y_train)
    train_score = model.score(x_train, y_train)
    test_score = model.score(x_test, y_test)

    if test_score > train_score:
        print("Test {} Train {} Random State {}".format(
            test_score, train_score, i
        ))

x_train, x_test, y_train, y_test = train_test_split(
    features, label, test_size=0.2, random_state=42
)

finalModel = LogisticRegression()
finalModel.fit(x_train, y_train)

print(finalModel)
print(finalModel.score(x_train, y_train))
print(finalModel.score(x_test, y_test))

print(classification_report(label, finalModel.predict(features)))
```

```
      User ID  Gender  Age  EstimatedSalary  Purchased
0    15624510    Male   19            19000          0
1    15810944    Male   35            20000          0
2    15668575  Female   26            43000          0
3    15603246  Female   27            57000          0
4    15804002    Male   19            76000          0
..        ...     ...  ...              ...        ...
395  15691863  Female   46            41000          1
396  15706071    Male   51            23000          1
397  15654296  Female   50            20000          1
398  15755018    Male   36            33000          0
399  15594041  Female   49            36000          1

[400 rows x 5 columns]
    User ID  Gender  Age  EstimatedSalary  Purchased
0  15624510    Male   19            19000          0
1  15810944    Male   35            20000          0
2  15668575  Female   26            43000          0
3  15603246  Female   27            57000          0
4  15804002    Male   19            76000          0
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Test 0.9 Train 0.840625 Random State 4
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Test 0.8625 Train 0.85 Random State 5
Test 0.8625 Train 0.859375 Random State 6
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Test 0.8625 Train 0.8375 Random State 9
Test 0.9 Train 0.840625 Random State 10
Test 0.8625 Train 0.85625 Random State 14
Test 0.85 Train 0.84375 Random State 15
Test 0.8625 Train 0.85625 Random State 16
Test 0.875 Train 0.834375 Random State 18
Test 0.85 Train 0.84375 Random State 19
Test 0.875 Train 0.84375 Random State 20
Test 0.8625 Train 0.834375 Random State 21
Test 0.875 Train 0.840625 Random State 22
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Test 0.85 Train 0.834375 Random State 26
Test 0.85 Train 0.840625 Random State 27
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Test 0.8875 Train 0.840625 Random State 38
Test 0.875 Train 0.8375 Random State 39
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Test 0.875 Train 0.846875 Random State 46
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Test 0.9 Train 0.84375 Random State 54
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Test 0.925 Train 0.8375 Random State 61
Test 0.8875 Train 0.834375 Random State 65
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Test 0.9 Train 0.83125 Random State 72
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Test 0.9 Train 0.825 Random State 106
Test 0.8625 Train 0.840625 Random State 107
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Test 0.85 Train 0.840625 Random State 111
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Test 0.9125 Train 0.840625 Random State 112
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Test 0.8625 Train 0.840625 Random State 116
Test 0.875 Train 0.834375 Random State 119
Test 0.9125 Train 0.828125 Random State 120
Test 0.8625 Train 0.859375 Random State 125
Test 0.85 Train 0.846875 Random State 128
Test 0.875 Train 0.85 Random State 130
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Test 0.8875 Train 0.83125 Random State 206
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Test 0.85 Train 0.84375 Random State 212
Test 0.8625 Train 0.834375 Random State 214
Test 0.875 Train 0.83125 Random State 217
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Test 0.8625 Train 0.84375 Random State 346
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Test 0.8625 Train 0.85 Random State 356
Test 0.9125 Train 0.840625 Random State 357
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Test 0.875 Train 0.828125 Random State 388
Test 0.85 Train 0.84375 Random State 394
Test 0.8625 Train 0.8375 Random State 395
Test 0.9 Train 0.84375 Random State 397
Test 0.8625 Train 0.84375 Random State 400
LogisticRegression()
0.8375
0.8875
              precision    recall  f1-score   support

           0       0.85      0.93      0.89       257
           1       0.85      0.70      0.77       143

    accuracy                           0.85       400
   macro avg       0.85      0.81      0.83       400
weighted avg       0.85      0.85      0.84       400
```